

1

The Four Purposes of Multidimensional Scaling

Multidimensional scaling (MDS) is a method that represents measurements of similarity (or dissimilarity) among pairs of objects as distances between points of a low-dimensional multidimensional space. The data, for example, may be correlations among intelligence tests, and the MDS representation is a plane that shows the tests as points that are closer together the more positively the tests are correlated. The graphical display of the correlations provided by MDS enables the data analyst to literally “look” at the data and to explore their structure visually. This often shows regularities that remain hidden when studying arrays of numbers. Another application of MDS is to use some of its mathematics as models for dissimilarity judgments. For example, given two objects of interest, one may explain their perceived dissimilarity as the result of a mental arithmetic that mimics the distance formula. According to this model, the mind generates an impression of dissimilarity by adding up the perceived differences of the two objects over their properties.

In the following, we describe four purposes of MDS: (a) MDS as a method that represents (dis)similarity data as distances in a low-dimensional space in order to make these data accessible to visual inspection and exploration; (b) MDS as a technique that allows one to test if and how certain criteria by which one can distinguish among different objects of interest are mirrored in corresponding empirical differences of these objects; (c) MDS as a data-analytic approach that allows one to discover the dimensions that underlie judgments of (dis)similarity; (d) MDS as a psychological model that explains judgments of dissimilarity in terms of a rule that mimics a particular type of distance function.

TABLE 1.1. Correlations of crime rates over 50 U.S. states.

Crime	No.	1	2	3	4	5	6	7
Murder	1	1.00	0.52	0.34	0.81	0.28	0.06	0.11
Rape	2	0.52	1.00	0.55	0.70	0.68	0.60	0.44
Robbery	3	0.34	0.55	1.00	0.56	0.62	0.44	0.62
Assault	4	0.81	0.70	0.56	1.00	0.52	0.32	0.33
Burglary	5	0.28	0.68	0.62	0.52	1.00	0.80	0.70
Larceny	6	0.06	0.60	0.44	0.32	0.80	1.00	0.55
Auto theft	7	0.11	0.44	0.62	0.33	0.70	0.55	1.00

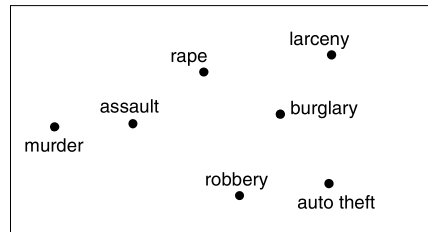


FIGURE 1.1. A two-dimensional MDS representation of the correlations in Table 1.1.

1.1 MDS as an Exploratory Technique

Exploratory data analysis is used for studying theoretically amorphous data, that is, data that are not linked to an explicit theory that predicts their magnitudes or patterns. The purpose of such explorations is to help the researcher to *see* structure in the data. MDS, too, can be used for such data explorations.

Consider an example. The U.S. Statistical Abstract 1970 issued by the Bureau of the Census provides statistics on the rate of different crimes in the 50 U.S. states (Wilkinson, 1990). One question that can be asked about these data is to what extent can one predict a high crime rate of murder, say, by knowing that the crime rate of burglary is high. A partial answer to this question is provided by computing the correlations of the crime rates over the 50 U.S. states (Table 1.1). But even in such a fairly small correlation matrix, it is not easy to understand the structure of these coefficients. This task is made much simpler by representing the correlations in the form of a “picture” (Figure 1.1). The picture is a two-dimensional MDS representation where each crime is shown as a point. The points are arranged in such a way that their distances correspond to the correlations. That is, two points are close together (such as murder and assault) if their corresponding crime rates are highly correlated. Conversely, two points are far apart if their crime rates are not correlated that highly (such as assault and larceny). The correspondence of data and distances is tight in this

example: the product-moment correlation between the coefficients in Table 1.1 and the distances in Figure 1.1 is $r = -.98$.

The reader need not be concerned, at this point, with the question of how such an MDS representation, $\mathbf{X}$, is found. We return to this issue in considerable detail in later chapters. For now, it suffices to assume that the data are fed to an MDS computer program and that this program provides a best-possible solution in a space with a dimensionality selected in advance by the user. The quality of this solution can be checked without knowing how it was found. All one has to do is measure the distances between the points of $\mathbf{X}$ and compare them with the data.¹ If distances and data are highly correlated in the sense of the usual product-moment correlation, say, then the distances represent the data well in a linear sense.² This is obviously true in the given case, and so the distances in Figure 1.1 represent the correlations in Table 1.1 very precisely.

What does the MDS picture in Figure 1.1 tell us? It shows that the crimes are primarily distributed along a horizontal dimension that could be interpreted as “violence vs. property” crimes. Moreover, the “property crimes” are less homogeneous, exhibiting some spread along the vertical axis, a dimension that could be interpreted as “hidden vs. street” crimes.

Although here we looked at dimensions, it is important to keep in mind that *any* property of the MDS representation that appears unlikely to result from chance can be interesting. The points may, for example, form certain groupings or clusters. Or, they may fall into different *regions* such as a center region surrounded with bands. The points may also lie on certain *manifolds* such as curved lines (a circle, for example) or on some surface in a higher-dimensional space. Looking for particular directions that would explain the points’ distribution is just one possibility to search for structure. Later on in this book, we explore a variety of geometric regularities that have been found useful in practical research.

¹Consider an analogy. Anyone can check the proposition that the number 1.414 approximates $\sqrt{2}$ simply by multiplying 1.414 by itself. The result shows that the proposition is nearly correct. For checking it, it is irrelevant how the number 1.414 was found. Indeed, few would know how to actually compute such a solution, except by trial and error, or by pushing a button on a calculator.

²With few points, one can even do (two-dimensional) MDS by hand. To find an MDS solution for the data in Table 1.1, first cut out seven small pieces of paper and write onto each of them one of the labels of the variables in Table 1.1, i.e., “murder”, “rape”, ..., “auto theft”, respectively. Place these pieces of paper arbitrarily in a plane and then move them around in small steps so that higher correlations tend to correspond to smaller distances. Repeat these corrective point movements a few times until the match of distances and data is satisfactory or until it cannot be improved anymore. Such a manual approach is typically quite easy to perform as long as the number of variables is small. With many variables, computer algorithms are needed for doing the work. Good algorithms also make it more likely that one ends up with an optimal MDS solution, that is, a configuration whose distances represent the given data “best” (in some well-defined sense).

Such insights into the data structure are aided by the visual access made possible by the simple MDS picture. Of course, as it is true for exploratory data analysis in general, it is left to further studies to test whether the patterns thus detected are stable ones. Moreover, it is desirable to also develop a theory that provides a rationale for the findings and enables one to predict such structures.

1.2 MDS for Testing Structural Hypotheses

When more is known about a field of interest, exploratory methods become less important. The research items, then, are well designed and the general interest lies in studying effect hypotheses. That is, in particular, what one wants to know is if and how the facets (dimensions, factors, features, etc.) by which the items are conceptually distinguished are reflected in corresponding differences among observations on these items. MDS may be useful for studying such questions. Consider a case.

Levy (1983) reports a study on attitudes towards political protest behavior. She distinguished 18 types of attitudes towards political protest acts. These types correspond to the $3 \cdot 3 \cdot 2 = 18$ different ways of reading the following design scheme (mapping sentence):

$$\begin{array}{l}
 \text{The } \left\{ \begin{array}{l} \text{A: modality of attitude} \\ a_1 = \text{evaluation} \\ a_2 = \text{approval} \\ a_3 = \text{likelihood of own overt action} \end{array} \right\} \text{ behavior of respondent } x \\
 \\
 \text{with respect to } \left\{ \begin{array}{l} \text{B: strength of execution} \\ b_1 = \text{demanding} \\ b_2 = \text{obstructive} \\ b_3 = \text{physically damaging} \end{array} \right\} \text{ protest acts of} \\
 \\
 \left\{ \begin{array}{l} \text{C: way to carry out} \\ c_1 = \text{omission} \\ c_2 = \text{commission} \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \text{R: direction} \\ \text{very positive} \\ \text{to} \\ \text{very negative} \end{array} \right\} \text{ behavior towards acts.}
 \end{array}$$

Thirty items were selected from a study by Barnes et al. (1979), using this mapping sentence as a culling rule. Short verbal labels and the codings for the selected items with respect to the three facets of the mapping sentence are given in Table 1.2. For example, item no. 6 effectively asked: “To what extent is ‘painting slogans on walls’ effective when people use this act in pressing for change?” The respondent’s answer was, for this item, recorded on a scale from “very effective” to “not effective”. (This scale is the “range” R of the observational mapping.) According to Levy, this item asks about